Guanyu **Mi**

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Education

University of California, Davis — *Davis, CA*
Master of Science in Electrical and Computer Engineering — *Sep 2024 – Jun 2026*

- **Coursework:** Large Foundation Models and Personalization, Foundations of Large Language Models, Unsupervised Learning, Practical AI, Random Signals & Noise, VLSI Digitial Signal Processing

The Chinese University of Hong Kong, Shenzhen — *Shenzhen, China*
Bachelor of Engineering in Electronic Information Engineering — *Sep 2020 – May 2024*

- **Coursework:** Machine Intelligence and Applications, Image Processing and Computer Vision, Digital Signal Processing, Microprocessors and Computer Systems, Data Structures, Optimization

Skills

Languages	Python, C/C++, JavaScript/TypeScript
AI & ML Frameworks	PyTorch, TensorFlow, CUDA, Scikit-learn, Pandas, NumPy, SciPy
LLM & Agentic tools	vLLM, Unsloth, LangGraph, veRL, TRL, OpenAI/Gemini APIs, FastAPI
Developer Tools	Linux, Docker, Git, Github Copilot, Codex

Experience

Research Assistant — *Davis, CA*
Self-Supervised Learning (SSL) Lab, Advised by Prof. Yubei Chen — *Jun 2025 – Present*

- Extending the **Search-R1** research on multi-hop search to explore reasoning limits for an agentic **4B-parameter LLM** on the MuSiQue benchmark.
- Built a data synthesis pipeline using the **Gemini API** to construct golden trajectories; achieved a **2.7x exact match increase** (from 0.090 to 0.243) after optimizing tool-calling and cold-start SFT.
- Conducting ablation studies on **pass@k scaling laws** to evaluate reasoning performance as generation attempts scale, establishing upper bounds for search capabilities.
- Architecting a multi-turn reinforcement learning framework based on **veRL** to further optimize reasoning and retrieval policies.

Engineer Intern — *Shenzhen, China*
National Innovation Center for Advanced Medical Devices — *Jun 2023 – Aug 2023*

- Engineered real-time **C++** firmware for resource-constrained MCUs, optimizing a low-latency data pipeline to process continuous 9-axis IMU streams and extract complex kinematic features.
- Architected an end-to-end data acquisition system featuring low-power **BLE** transmission and a custom **PyQt** application for real-time multi-channel sensor parsing, synchronization, and visualization.

Projects

Adaptive Mock Interview Plugin — *Open-Source Extension @ personal*
https://github.com/GuanyuMi/Resume-Matcher-with-mock-interview-plugin — *Feb 2026*

- Developing a Modular Monolith extension for the open-source Resume-Matcher system, embedding custom **FastAPI** routers to deliver a native, interactive mock-interview experience.
- Engineering a dynamic GenAI pipeline using **LangChain** and structured outputs to generate context-aware technical questions based on resume-JD gap analysis, effectively eliminating LLM hallucinations.
- Designing an **SQLite** data pipeline to track user performance metrics, which feeds into a **scikit-learn** predictive model being trained to dynamically adjust question difficulty based on individual learning curves.

Efficient Reproduction of Logic-RL with Unsloth — *Research Project @ SSL Lab*
https://github.com/GuanyuMi/logic-rl-unsloth-repro — *Jul 2025*

- Reproduced the key experiments from the **Logic-RL** paper by assembling a fine-tuning workflow using existing tools such as **Unsloth**, **LoRA**, **TRL**, and **vLLM**, enabling DeepSeek-R1-style reasoning training on a single 16GB V100 GPU.
- Performed systematic hyperparameter tuning through **TensorBoard**-based analysis, adjusting KL-penalty weights and reward configurations to identify a locally optimal setup under limited computing resources.